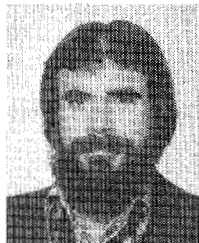


Francis A. Reed (M'79) was born in New Haven, CT, on May 20, 1944. He received the B.S.E.E. degree from the University of Connecticut, Storrs, in 1966, and the M.S.E.E. degree in communication sciences from the University of Southern California, Los Angeles, in 1978, and has done graduate work in communication theory at Northeastern University, Boston, MA.

Since 1976, he has been with Hughes Aircraft Company, Fullerton, CA, working in the areas of statistical classification, adaptive processing, detection, and estimation as applied to sonar systems. Previous work has also included the application of digital signal processing to acoustic signature simulation and the system design of large scale digital sonars. He is presently working toward the Ph.D. degree in electrical engineering at the University of Southern California under a Hughes Aircraft Company fellowship.



Paul L. Feintuch (S'66-M'67) received the B.S.E.E. degree from the Cooper Union, New York, NY, in 1967, and the M.S.E.E. and Ph.D. degrees in electrical engineering from the University of Southern California, Los Angeles, in 1969 and 1972, respectively.

He is currently with the Hughes Aircraft Company, Fullerton, CA, where he is Senior Scientist in the Systems Laboratory, with primary interest in research and development in all aspects of sonar signal processing and in the

applications of adaptive filtering to sonar systems. He is also an Adjunct Assistant Professor of Electrical Engineering at the University of Southern California.

Dr. Feintuch is past Chairman of the Los Angeles Chapter of the IEEE Professional Group on Information Theory.



Neil J. Bershad (S'60-M'62) was born in Brooklyn, NY, on October 20, 1937. He received the B.E.E. degree from Rensselaer Polytechnic Institute, Troy, NY, in 1958, the M.S. degree in electrical engineering from the University of Southern California, Los Angeles, in 1960, and the Ph.D. degree in electrical engineering from Rensselaer Polytechnic Institute in 1962.

In 1966 he joined the Faculty of the School of Engineering, University of California, Irvine, where he is now a Professor of Electrical Engineering. His research interests have involved stochastic system modeling and analysis. His present interests are in the areas of detection and estimation using multiple observations, and the statistical behavior of nonlinear systems driven by stochastic inputs. For the past five years, he has served as an Associate Editor of the IEEE TRANSACTIONS ON COMMUNICATIONS in the area of phase-locked loops and synchronization. He is presently a Consultant in array processing to the Hughes Aircraft Company, Fullerton, CA.

Time Delay Estimation Using the LMS Adaptive Filter—Dynamic Behavior

PAUL L. FEINTUCH, MEMBER, IEEE, NEIL J. BERSHAD, MEMBER, IEEE, AND FRANCIS A. REED, MEMBER, IEEE

Abstract—The LMS adaptive filter is used to estimate a linearly moving time delay between two broad-band waveforms. The tracking behavior of the mean weights is analyzed and is compared with simulations of the actual device.

I. INTRODUCTION

THE concept of using an *LMS adaptive filter*, configured as a canceller to estimate the time delay difference between two waveforms, is studied in [1] for a static time delay. The estimate of delay is obtained by interpolating on the weights in the filter to select the point in the tapped delay line that

corresponds to the peak weight. The resulting estimate is unbiased in the nonmoving input case, and the variance is within 0.5 dB of the Cramér-Rao lower bound for the variance of any delay estimate using the same data. In addition, it was shown that if assumed *a priori* statistics concerning the input power spectra differ from what is actually the case, then the adaptive filter tracker can dramatically outperform a fixed parameter conventional tracker being operated in an environment other than that for which it was designed.

Since the *delay estimation* is performed on the peak value of the weights in the adaptive filter, the time constant for the selection of the peak is the significant transient response. The peak weight can be correctly selected much more quickly than the time required for either the weights or the mean square error to converge [1]. This suggests that this processor has the potential to track rapidly changing time delays by just observing the peak weight move through the adaptive filter tapped delay line.

Manuscript received May 9, 1980; revised September 30, 1980. This work was supported by the Naval Sea Systems Command Code 63R under Contract N00024-77-C-6251.

P. L. Feintuch and F. A. Reed are with Hughes Aircraft Company, Fullerton, CA 92634.

N. J. Bershad is with the Department of Electrical Engineering, University of California, Irvine, CA 92664, and is a consultant to Hughes Aircraft Company, Fullerton, CA 92634.

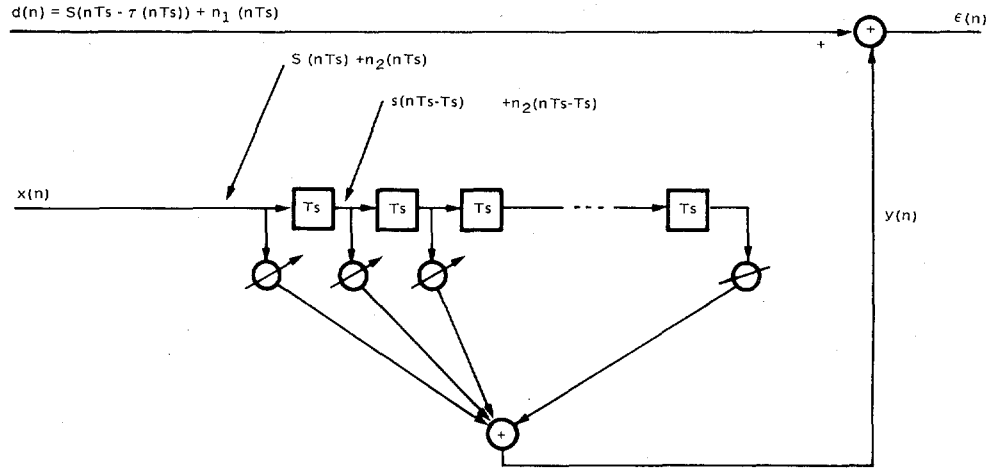


Fig. 1. Data in the delay line.

This paper extends the results of [1] to consider a *linearly time-varying delay*. The *mean weights* are derived for both spectrally white signals and band-limited broad-band signals with exponential correlation function and with small signal-to-noise ratio. The mean weights in steady state are derived and shown to be time varying and lagging the ramp input. Simulations are then presented, demonstrating good agreement with the analytical results.

II. ANALYSIS

A. Mean Weight Behavior of Adaptive Filter Tracker

In this section, the time-varying mean weights of the adaptive filter are derived for a linearly time-varying change of delay. Two signal models are considered—a broad-band spectrally white signal and a broad-band nonwhite signal.

For a spectrally white process, the mean filter weights are shown to be a traveling wave with a decaying exponential envelope. Tracking the time delay involves estimating the delay location of the leading edge of the weights. For the broad-band nonwhite signal, the mean weights behave similarly except, because of correlation between the taps, the peak of the traveling wave is much larger and the decaying exponential envelope is dependent on the signal dynamics and correlation.

1) *Mean Weights for Broad-Band White Signals in Uncorrelated Noise*: Let the time delay between the two input channels be given by $D(t) = bt$. The input signal and noise are assumed to be independent zero-mean Gaussian random processes that are spectrally white over the band corresponding to the sampling frequency, with power σ_s^2 and σ_n^2 , respectively. One channel, $d(n)$, is the desired signal for the adaptive process, and the other is the input $x(n)$ to the adaptive filter.

The algorithm for changing the weights in the adaptive filter is given by [2]

$$\begin{aligned} W(n+1) &= W(n) + \mu [d(n) - X^T(n) W(n)] X(n) \\ &= W(n) + \mu [d(n) X(n) - X(n) X^T(n) W(n)] \end{aligned} \quad (1)$$

where

$W(n)$ = filter weight vector at time sample n
 $d(n)$ = desired signal at time sample n
 $X(n)$ = observed data vector of samples
 μ = LMS algorithm step size.

T denotes vector transpose. Fig. 1 shows the data vector $X(n)$ in the tapped line. The scalar $d(n)$ is the delayed and sampled signal plus noise,

$$d(n) = s(nT_s - D(nT_s)) + n_1(nT_s) \quad (2)$$

and the vector $X^T(n)$ is

$$X^T(n) = (s(nT_s), s((n-1)T_s), \dots, s((n-M)T_s)) + N_2^T(nT_s) \quad (3)$$

where

T_s = time delay between taps = algorithm sampling time
 M = number of taps.

It is first assumed that the signal is white, so that

$$E[s(n)s(m)] = \sigma_s^2 \delta(n-m) \quad (4)$$

where $\delta(n-m) = 1$ if $n=m$ and $\delta(n-m) = 0$ otherwise. Taking expectations of (1) and making the assumption [3] that the data and weights at time n are uncorrelated yields a difference equation for the mean of the adaptive filter weight,¹

$$E[W(n+1)] = [I - \mu R_{xx}(n)] E[W(n)] + \mu R_{dx}(n) \quad (5)$$

where

$I = M \times M$ identity matrix

$$R_{dx}(n) = E[d(n) X(n)]$$

$$R_{xx}(n) = E[X(n) X^T(n)].$$

Since $R_{xx}(n)$ is independent of n , and the initial mean weight can be $E[W(0)] = 0$, then (5) can be rewritten as

$$E[W(n)] = \mu \sum_{k=0}^{n-1} [I - \mu R_{xx}(k)]^{n-k-1} R_{dx}(k). \quad (6)$$

The second-order statistics can be calculated from the input waveforms as follows:

$$R_{xx}(n) = (\sigma_s^2 + \sigma_n^2) I \quad (7)$$

$$R_{dx}^T(n) = (\delta(bnT_s), \delta(bnT_s - T_s), \dots, \delta(bnT_s - MT_s)). \quad (8)$$

¹The time compression parameters, as discussed in [6] for example, are imbedded in the cross correlation properties of the taps and hence are contained directly in $R_{dx}(n)$ and $R_{xx}(n)$.

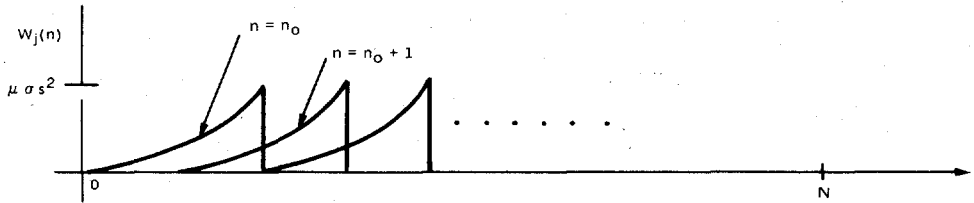


Fig. 2. Envelope of the tracker weights as they move in time.

Using (6), the mean weight vector at the n th iteration is given by

$$E[W(n)] = \mu \sigma_s^2 \sum_{k=0}^{n-1} [1 - \mu(\sigma_s^2 + \sigma_n^2)]^{n-k-1} R_{dx}(k). \quad (9)$$

In the static case, $b = 0$ and only the first weight is nonzero, with mean value

$$E[W_0(n)] = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_n^2} [1 - (1 - \mu(\sigma_s^2 + \sigma_n^2))^n] \quad (10)$$

which converges to the Wiener filter [4] for the broad-band stationary case as $n \rightarrow \infty$.

In the dynamic case, the weights are a moving set of spikes that are changing with amplitude as the signal moves and as the weights converge. The total weight vector is the sum of the vectors in (9). The weights can be viewed as a sliding window of exponentially growing responses, or as a moving weight at the leading edge that leaves behind it an exponentially decaying wake.

This can be seen by examining the weight at the leading edge of the response. In (9), the leading edge will occur at the latest time. If the filter is sufficiently long so that the response still falls within the tapped delay line, i.e., $M > n$, then the amplitude and location of the leading weight are found by examining the term in the summation for which $k = n - 1$. The amplitude of the leading edge is $\mu \sigma_s^2$ and its location is at tap number $b(n - 1)$. If $b = 1$ then the signal moves one tap per iteration; if the signal changes more slowly, then $b < 1$ and the leading edge moves more slowly than the iteration rate.

For the special case where $b = 1$ the weight vector in (9) can be readily expanded. Letting $r = 1 - \mu(\sigma_s^2 + \sigma_n^2)$,

$$E[W^T(n)] = \mu \sigma_s^2 (r^{n-1}, r^{n-2}, \dots, r^1, 1, 0, \dots) \quad (11)$$

which shows the decaying wake behind the leading weight which shifts along the delay line as n increases, as pictured in Fig. 2.

For $b < 1$ the weights move more slowly and the basic model herein tends to become less realistic. The signal sequence used is uncorrelated in time. In general this is not the case. The impact of this assumption is to have weights respond only at the exact correct alignment of input delays and tap delay values. In the band-limited, nonwhite signal case, correlation will exist even at noninteger delay shifts, and larger weight responses should be expected at the leading edge for slower moving signals. It is shown in the next section that the amplitude of the leading edge decreases monotonically with b from $\sigma_s^2/(\sigma_s^2 + \sigma_n^2)$ (the value for $b = 0$) to $\mu \sigma_s^2$ (the value for which movement is so fast that the signal samples essentially decorrelate totally at each time sample). The extent of the

wake and the height of the leading edge will depend on signal dynamics.

2) *Mean Weights for Band-Limited Broad-Band (Correlated) Signal in Uncorrelated Noise for Small Signal-to-Noise Ratios:* The conditions of the previous section hold with one extension. The signal spectrum is no longer white. Equations (2) and (3) are still valid. However, $R_{dx}(n)$ and $R_{xx}(n)$ [(7) and (8)] are now replaced by

$$\begin{aligned} R_{dx}^T(n) &= (R_s(bnT_s), R_s(bnT_s - T_s) \cdots R_s(bnT_s - MT_s)) \\ &= \sigma_s^2 (\rho^{|bnT_s|}, \rho^{|bnT_s - T_s|} \cdots \rho^{|bnT_s - MT_s|}) \\ &= \sigma_s^2 V^T(n) \end{aligned} \quad (12)$$

$$R_{xx}(n) = \sigma_n^2 I + \begin{bmatrix} \sigma_s^2 R_s(T_s) \cdots R_s(MT_s) \\ \text{SYM} \end{bmatrix} = \sigma_n^2 I + \sigma_s^2 \Lambda \quad (13)$$

where

$$\Lambda = \begin{bmatrix} 1 & \rho^{T_s} & \rho^{2T_s} & \cdots & \rho^{MT_s} \\ \text{SYM} \end{bmatrix}. \quad (14)$$

Since $R_{xx}(n)$ is independent of n , for $E[W(0)] = 0$, (6) simplifies to

$$E[W(n)] = \mu \sigma_s^2 \sum_{k=0}^{n-1} R^{n-k-1} V(k) \quad (15)$$

where

$$R = (1 - \mu \sigma_n^2) I - \mu \sigma_s^2 \Lambda. \quad (16)$$

In order to proceed further, it is necessary to assume small signal-to-noise ratio, i.e., $\sigma_s^2 \ll \sigma_n^2$. Then $R_{xx}(n) \approx \sigma_n^2 I$, $R \approx (1 - \mu \sigma_n^2) I$. The mean weight vector, for small signal-to-noise ratio, is approximately

$$E[W(n)] \approx \mu \sigma_s^2 \sum_{k=0}^{n-1} (1 - \mu \sigma_n^2)^{n-k-1} V(k). \quad (17)$$

The j th weight becomes

$$E[W_j(n)] = \mu \sigma_s^2 \sum_{k=0}^{n-1} (1 - \mu \sigma_n^2)^{n-k-1} \rho^{|bk-j|T_s}. \quad (18)$$

Assume that the filter is long enough so that $M = n - 1$, and let $j = Mb$ so that we are at the end of the filter. For $r = 1 - \mu \sigma_n^2$,

$$E[W_{Mb}(n)] = \mu \sigma_s^2 r^M \sum_{k=0}^M r^{-k} \rho^{-(k-M)bT_s} \\ = \mu \sigma_s^2 \frac{r^M \rho^{(M+1)bT_s} - 1}{r \rho^{bT_s} - 1}. \quad (19)$$

Equation (19) describes the transient behavior of the mean weight for small signal-to-noise ratios. In steady state, i.e., M arbitrarily large, one can readily see the relationships between signal correlation ρ , signal dynamics b , and algorithm dynamics μ :

$$\lim_{M \rightarrow \infty} E[W_{Mb}(\infty)] = \frac{\mu \sigma_s^2}{1 - r \rho^{bT_s}} = \frac{\mu \sigma_s^2}{1 - [1 - \mu \sigma_n^2] \rho^{bT_s}}. \quad (20)$$

There are several cases of interest. If $\rho = 1$, then the signal is totally correlated from tap to tap, regardless of signal dynamics. For this case $E[W_{Mb}(\infty)] = \sigma_s^2 / \sigma_n^2$. This is a small signal-to-noise ratio approximation to $\sigma_s^2 / (\sigma_n^2 + \sigma_n^2)$ which is what one would expect in the static case. If $\rho = 0$, the signal is uncorrelated and again signal dynamics should not affect the result. For this case, $E[W_{Mb}(\infty)] = \mu \sigma_s^2$ which agrees with (11) where dynamics tended to decorrelate the signal. If there are no signal dynamics, i.e., $b = 0$, then $E[W_{Mb}(\infty)] = \sigma_s^2 / \sigma_n^2$ independent of ρ , which is again the expected small signal-to-noise ratio static result. The tradeoff between dynamics and signal correlation can be seen by examining the term ρ^{bT_s} in the expression for $E[W_{Mb}(\infty)]$. As b decreases, ρ^{bT_s} looks more like unity and the signal looks more correlated. As b increases, ρ (which is less than one) is raised to a higher power and the signal has become less correlated.

B. Steady-State Mean Weights

Equation (18) is an expression for the mean weights of the adaptive tracker, for small signal-to-noise ratios, when the input signal is a broad-band correlated process with linearly time-varying delay, buried in uncorrelated noise. Equation (18) can be expressed, in steady state, in closed form as [5, Appendix G]

$$\lim_{n \rightarrow \infty} E[W_j(n)] = \mu \sigma_s^2 \left[\frac{(1 - \mu \sigma_n^2)^{n-l-1}}{1 - (1 - \mu \sigma_n^2) \rho^{bT_s}} \right. \\ \left. + \frac{\rho^{bT_s(n-l-1)} - (1 - \mu \sigma_n^2)^{n-l-1}}{1 - (1 - \mu \sigma_n^2) \rho^{-bT_s}} \right], \\ j \leq b(n-1) \\ = \frac{\mu \sigma_s^2}{1 - (1 - \mu \sigma_n^2) \rho^{bT_s}} \rho^{-bT_s[n-l-1]}, \\ j \geq b(n-1) \quad (21)$$

where

$$l = \text{integer such that } l \leq \frac{j}{b} < l+1, \text{ and } |j - bl|T_s \ll 1.$$

The weights have an exponentially decaying leading and trailing edge. For $\mu \sigma_n^2 < 1 - \rho^{bT_s}$, the trailing edge of the weights has a decay factor $1 - \mu \sigma_n^2$. For $\mu \sigma_n^2 > 1 - \rho^{bT_s}$, the

trailing edge of the weights has a decay factor ρ^{bT_s} . The leading edge of the weights has decay factor ρ^{bT_s} independent of the ratio. Hence, for $\mu \sigma_n^2 > 1 - \rho^{bT_s}$, the shape of the trailing edge of the weights is determined by the input signal dynamics, whereas for $\mu \sigma_n^2 < 1 - \rho^{bT_s}$, the shape of the trailing edge of the weights is determined by the system dynamics.

The lag in the peak of the weights in (21) in comparison to the true delay is given by

$$\text{tap lag} = \left[\frac{b}{b\alpha T_s + \ln(1 - \mu \sigma_n^2)} \right] \\ \cdot \ln \left\{ \frac{\alpha b T_s [1 - (1 - \mu \sigma_n^2) \rho^{bT_s}]}{-\ln(1 - \mu \sigma_n^2) [(1 - \mu \sigma_n^2) \rho^{-bT_s} - \rho^{bT_s}]} \right\} \quad (22)$$

$$\approx \frac{b}{b\alpha T_s - \mu \sigma_n^2} \ln \left[\frac{1}{2} + \frac{b\alpha T_s}{2\mu \sigma_n^2} \right] \quad (23)$$

where $\rho = e^{-\alpha}$. Thus the lag increases with the rate of the signal b , and decreases with larger feedback coefficient, μ .

III. SIMULATIONS

Simulations were run of a 16-tap adaptive filter operating at a sample rate 2400 Hz with broad-band (i.e., spectrally flat) inputs band-limited to 800 Hz, and a delay between signals that is linearly varying with time. The band-limited inputs were generated by passing computer generated white noise through a digital FIR filter. In Section II, it was shown that the mean weight vector, as a function of time, is a peak that follows the instantaneous delay between array halves, moving through the filter at the delay rate of change with lag and with an exponentially decaying trailing edge. This analysis is verified by Fig. 3, which shows the weight vector at 5000, 6000, and 7000 iterations for a 10 dB signal-to-noise ratio (SNR) and a delay changing at 1.0 ms/s.

A more quantitative assessment of the tracking behavior of the *adaptive tracker* can be had from Fig. 4, showing the delay estimate as a function of time for a linearly increasing delay of 261.8 $\mu\text{s/s}$, with $\mu = 2^{-10}$ for an SNR of 0 dB and $\mu = 2^{-14}$ for an SNR of -10 dB. Note that the delay estimate lags the time but has the correct time-averaged rate of change. The losses of track in Fig. 4(b) occur when weight fluctuations obscure the peak of the mean weights. The instantaneous estimates of delay while tracking will contain two sources of error—dynamics and random noise. The noise errors will cause the instantaneous delay estimates to deviate from a straight line. The dynamic errors will cause the estimate to lag. If one time averages with a moving window, the random noise errors will be reduced (although not dramatically because the fluctuations are probably highly correlated—the LMS algorithm has already performed significant averaging). The remaining error will be due to a fixed lag which is generally supported by Fig. 4. Evaluating the lag using (23) for this example produces a value of 0.0008 s; which is greater than the average lag of 0.003 s in Fig. 4(b). This difference may be due to the different spectral shape in the analysis and the simulations, causing the bandwidth parameters to disagree.

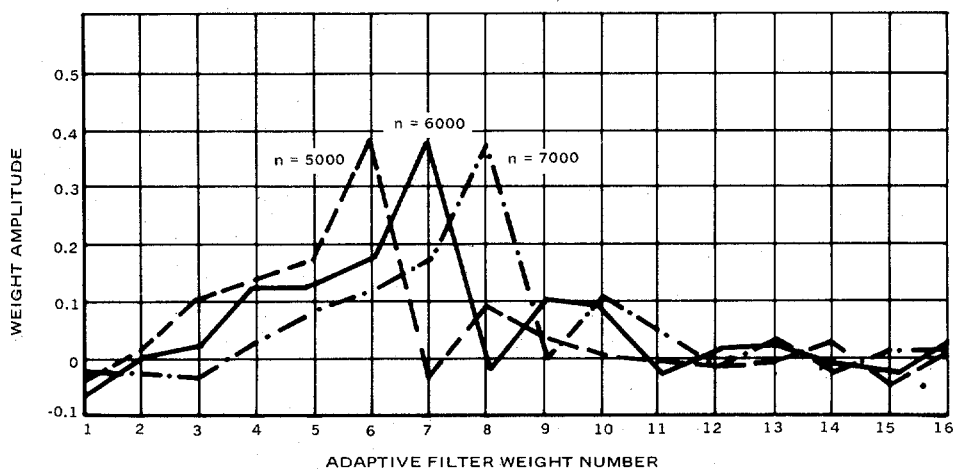


Fig. 3. Weight vector as a function of time (n) for linearly varying delay and broad-band input, N = number of taps = 16, SNR = 10 dB, $\mu = 2^{-10}$, $b = 1.0$ ms/s.

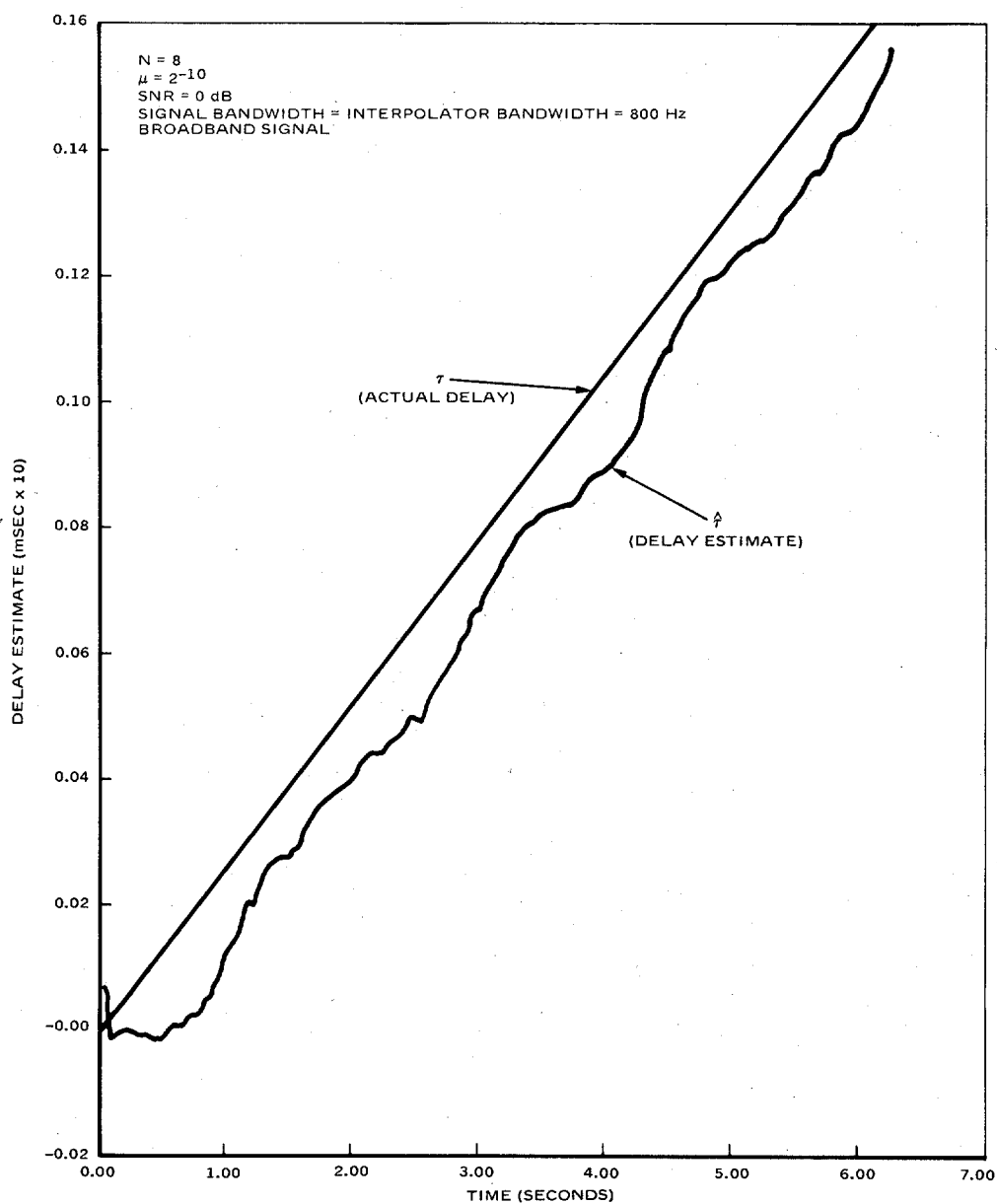


Fig. 4. (a) Delay estimate versus time for linearly varying delay ($261.8 \mu\text{s/s}$).

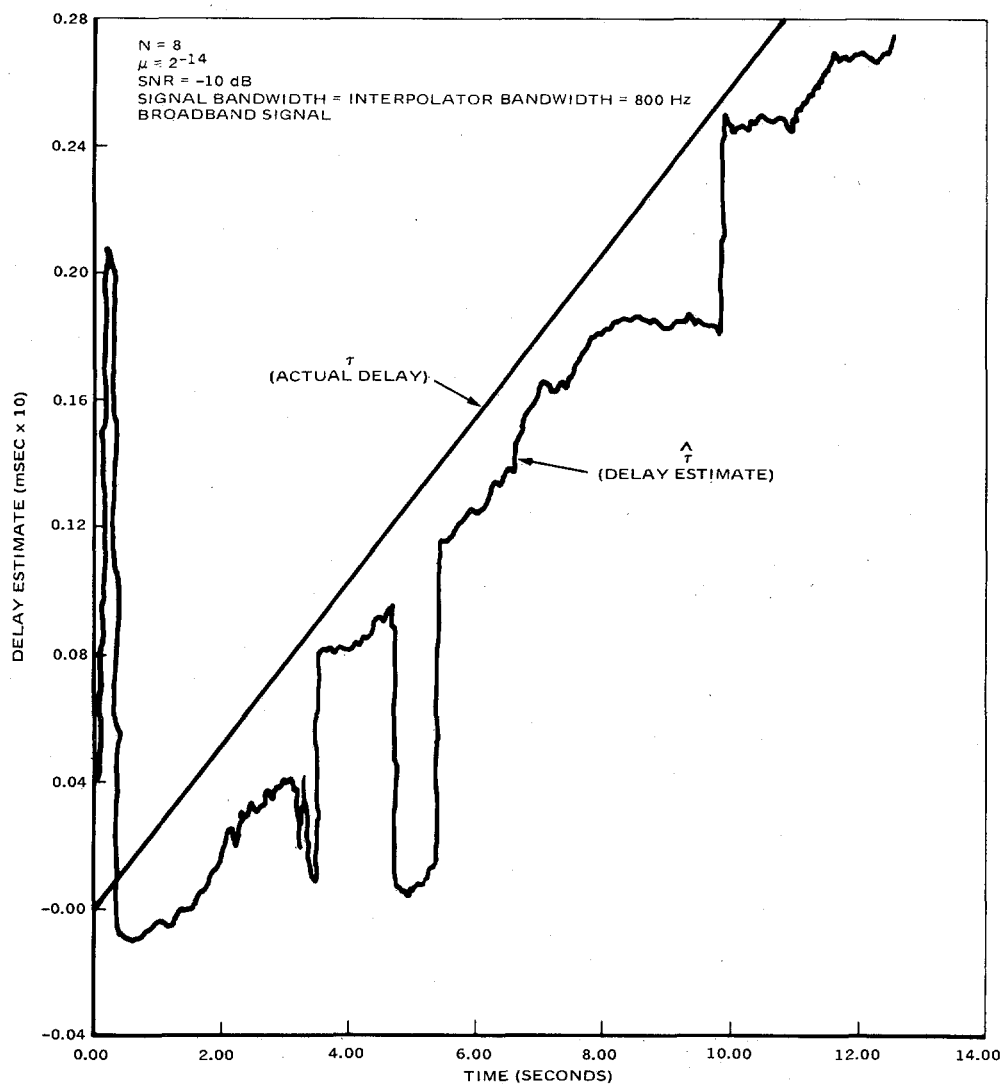


Fig. 4 (Continued). (b) Delay estimate versus time for linearly varying delay ($261.8 \mu\text{s/s}$).

IV. CONCLUSIONS

The transient and steady tracking behavior of the mean value of the LMS adaptive tracker weights have been derived for both spectrally white and nonwhite signal processes with linearly time-varying delays. The mean filter weights were shown to be a traveling wave which leaves behind an exponentially decreasing wake. It was shown that the trailing edge of the adaptive filter mean weight is determined by the algorithm dynamics for small weight update adjustments and by the input signal dynamics for large weight update adjustments.

It was also shown that the location of the peak of the weights lags the true time delay by an amount that depends on the delay rate, the signal correlation function, and the adaptive filter time response. For a large adaptive filter time constant, the lag is linear with respect to delay rate and inversely proportional to the adaptive filter time constant.

REFERENCES

- [1] F. A. Reed, P. L. Feintuch and N. J. Bershad, "Time delay estimation using the LMS adaptive filter—Static behavior," this issue, pp. 561-571.
- [2] B. Widrow *et al.*, "Adaptive noise cancelling: Principles and applications," *Proc. IEEE*, vol. 63, pp. 1692-1716, Dec. 1975.
- [3] J. R. Treichler, "Transient and convergent behavior of the adaptive line enhancer," *IEEE Trans. Acoust., Speech, Signal Processing*, vol. ASSP-27, p. 53, Feb. 1979.
- [4] A. Papoulis, *Probability, Random Variables, and Stochastic Processing*. New York: McGraw-Hill, 1965.
- [5] P. L. Feintuch, F. A. Reed, N. J. Bershad, and C. M. Flynn, "Adaptive tracking system study—Phase 2," Hughes Aircraft Co., Fullerton, CA, Final Rep., Oct. 1979, prepared for NAVSEA Code 63R, Contract N00024-77-C-6251.
- [6] C. H. Knapp and G. C. Carter, "Estimation of time delay in the presence of source and receiver motion," *J. Acoust. Soc. Amer.*, vol. 61, June 1977.

Paul L. Feintuch (S'66-M'67), for a photograph and biography, see this issue, p. 571.

Neil J. Bershad (S'60-M'62), for a photograph and biography, see this issue, p. 571.

Francis A. Reed (M'79), for a photograph and biography, see this issue, p. 571.